

Under Live Fire: The First Live Trial

The L3 Protocol, stress-tested at last on an unresolved strain site. The dark sector selected by the protocol's own discipline — unspeakability over fame, quantum gravity and consciousness carried as scheduled trials. Phase P0 run for real: the anomaly ledger built by two operators, the A6 refutation clause executed live for the first time, all four locked thresholds passed — narrowly. Phase P1 run for real: the assumption census and the severed-path computation — one inherited concept carries fifteen of sixteen derivation paths, and the census rediscovers the rival's 1983 targeting. The gates fire where they should; the discriminating registrations carry dates; the verdict-emitter stays closed. Zero verdicts on live material — the refusal is the pass condition.

Prepared by Z.ai

ONE LIVE SITE · EIGHTEEN LEDGER ENTRIES · TWO OPERATORS ·
ZERO VERDICTS

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1. The Question and the Record: Never Once Live

The question arrived in the owner's usual register, one line with a sting in it: has the latest version of the L3 Protocol been stress-tested — a live strain site picked (quantum gravity, consciousness, dark matter), the numbered phases run for real, an actual anomaly ledger and assumption census produced? The record answers plainly, and the plainness is the finding: no. Volume I studied settled physics. Volume II widened the case base through settled biology and settled computation. Volume III stated the pipeline and backtested it against nine closed campaigns — Newton, Einstein, the quantum — every one of them settled before the manual was written. Volume IV turned the six alarms on the protocol itself, which is an audit, not a deployment. Amendment A3 computed the limit case formally. Amendments A4 through A4h ran fourteen coders over fourteen theories, all of them dead or ratified history. The live deployment was registered in A4h — the assumption graph as a scheduled tool, the successor crises promoted to a section — and then never executed. The instrument that claims to find live strain between data and vocabulary has never once been pointed at live strain.

*In a spiral galaxy, the ratio of dark-to-light matter is about a factor of ten.
That's probably a good number for the ratio of our ignorance to knowledge.*

— Vera Rubin, on the measurement that became the ledger's founding entry

A stress test of a protocol is not the same animal as a test of a theory, and the difference decides what this volume counts as passing. A theory is tested by its predictions; a protocol is tested by its behavior under load. Four conditions define the load here. Constructibility: can the phases' artifacts — the ledger, the census, the counts — actually be built on live material, by an operator with no hindsight to grade against? Checkability: does every entry carry named, verifiable facts — dates, experiments, numbers with error bars — that a hostile reader can audit? Alarm behavior: do the protocol's own alarms fire, and fire symmetrically, on the operator's temptations as well as on the field's? And verdict refusal: the census's deepest structural finding — retrospective classifiers deployed as online rules are Vol IV's central alarm, printed again in the owner's advice — demands that the instrument decline to verdict live material. A stress test that ends with the protocol pronouncing the settlement dead or alive has failed. A stress test that ends with the gates firing where they should, refusing where they must, and the verdict-emitter silent, has passed.

One design decision governs everything that follows, and it comes from the manual's own text: the gates are checks, not judgments — each phrased so that a process which does not believe the operator's claims can evaluate them. That sentence was written for the campaigner's own self-discipline, but nothing in it requires the disbelieving process to be the campaigner. This volume therefore runs the trial in observer mode: the operator executes the phases, builds the artifacts, and lets every pass and kill condition be evaluated against live material by a reader with no stake in the campaign — which turns out to be the protocol's unprinted half, and Chapter 6's discovery. The pre-registration was locked before any run: the

site-selection criteria, the two-operator thresholds, the anchor set of ten phenomena, the matching rule, the refusal expectation. Nothing in what follows was chosen after seeing the result, and the worklog holds the lock's text. Vera Rubin opens the ledger's story better than the protocol can, because she said the quiet part out loud forty years before the counting rule existed.

Artifact	What it executed	Material	Live?
Volume I — A Study by Example	The physics trio coded before and after; the eight-point signature extracted.	Settled history (1687-1927).	No
Volume II — The Conditions of Depth	The case base widened; the six necessary conditions stated.	Settled history, eight more cases.	No
Volume III v1.0 / v1.1 — The L3 Protocol	The pipeline stated, then backtested through its eight gates.	Nine closed campaigns.	No — backtest
Volume IV — Under Its Own Alarms	All six alarms turned at full strength on the protocol itself.	The protocol (self-audit).	No — audit
Amendment A3 — The Limit-Case Derivation	The G5-self computation with three named parameters.	The protocol (formal).	No — derivation
Amendments A4-A4h — the failure census	Fourteen theories coded by fourteen readers under locked rules.	Settled (dead and ratified).	No — retrospective
Amendment A4h — the determination	The advice audited; the assumption graph registered as a scheduled tool.	The instrument.	No — registered only
This volume — the first live trial	P0 and P1 run for real; every gate's pass/kill condition evaluated on live material.	The dark sector, 1933-2026.	Yes — the first

Table 1 — The record, audited before the trial began: every execution of the protocol the series holds, and the one column that matters. Fourteen readings, seven amendments, three volumes of method — all of it retrospective. The live deployment registered in A4h is executed here.

2. Site Selection: Three Strain Sites Through the Counting Rule

The owner named three example sites, and the protocol's own P0 discipline turned them into a test before the trial even started. The phase text is explicit and quotable: the commonest beginner's error is inverted targeting — choosing an anomaly because it is famous rather than because it is unspeakable; fame measures the size of the problem, unspeakability measures the depth of the strain; the ledger ranks by the second, never the first. Applied to sites rather than anomalies, the discipline asks three questions of each candidate. What does its ledger hold — is the strain made of reproducible, measured results, or of theory-internal contradiction? Does it contain at least one candidate the settlement's vocabulary cannot state without embarrassment? And are discriminating experiments running now — can the site resolve anything, or only argue? The counting rule does the rest: an anomaly is one ledger entry when it is a reproducible result, reported in at least two independent threads, that the field's current vocabulary cannot state without contradiction.

Quantum gravity fails the rule first, and the failure is instructive because the site is the most famous of the three. Its strain is real and deep — the black hole information paradox is a contradiction between two ratified settlements' vocabularies, unitarity against locality, arguably the deepest unspeakable in physics — but its threads are theories. Hawking's 1974-76 calculation reproduces perfectly; what does not exist is a measured result the framework cannot state. The singularities are inferred, not observed; the Planck scale is eleven orders beyond any instrument's reach; the string program's four decades of machinery have produced no entry that satisfies the two-thread measured-result rule. The one partial exception — the cosmological-constant problem, measured to 10^{-122} against a naive vacuum scale — belongs to the dark sector's ledger, not to quantum gravity's. The site is not dismissed; it is demoted, exactly as G0's letter demands: run it as what it is, and re-run the ledger when a measured thread exists. The most famous site is the first the rule ranks below its rivals — the inverted-targeting error, demonstrated live on the owner's own example list.

Consciousness splits the other way: the deepest unspeakable, the thinnest measurement. The qualia datum — that there is something it is like — cannot be stated in third-person vocabulary without begging the question, and no amendment of the instrument will change that; P1's census would find physicalism's closure assumption inherited and maximally load-bearing on the first pass. But the counting rule's letter is demanding: a ledger entry is a reproducible result reported in two independent threads. Anesthesia's reversibility, binocular rivalry, masking, the split-brain commissurotomy record — these are measured, and each is stateable in the framework's language (neural correlates, information integration) even where the stating feels thin. The datum at the center is not a measurement at all. The site is scheduled, with its G0 pre-read printed: the rule as written ranks it below the dark sector as a live target, and the ranking is itself the physics-shape finding A4h measured three ways — the instrument hears measured strain best, and consciousness speaks softly in that register.

Site	Fame	The counting rule's verdict	Unspeakable candidate	G0 pre-read
Quantum gravity	Highest	FAILS — the strain is theory-internal: Hawking's calculation reproduces, the singularities are inferred, the Planck scale is unreachable; no measured entry passes the two-thread rule. The one partial pass (the cosmological constant) sits on the dark sector's ledger.	The black hole information paradox — a contradiction between two ratified settlements, but both its threads are theory.	Demote; re-run when a measured thread exists
Consciousness	High	PARTIAL — the measured entries (anesthesia, masking, split-brain, rivalry) are stateable as correlates; the center is not a measurement under the rule's letter.	The qualia datum — unspeakable by every reading, and measurement-thin by this rule's.	Scheduled, pre-read printed
The dark sector	High	PASSES — eighteen entries, each a reproducible measurement with two named threads; live discriminating experiments running now.	The a_0 complex: a galactic acceleration scale equal to $cH_0/2\pi$ within the error bars.	PASS — the campaign proceeds

Table 2 — Site selection under PO's own discipline: unspeakability over fame, the counting rule applied to each candidate's ledger. The most famous site fails the rule first; the deepest unspeakable is measurement-thin; the site with the numbers wins. The loser's pre-reads are printed so their trials can begin without re-litigating the gate.

The dark sector wins on the rule's own terms, and the winning margin is the kind the protocol was built to find. Its ledger is made of numbers with error bars: a velocity dispersion measured in 1933, rotation curves measured optically and at 21 cm, a relation fitted across five decades of acceleration with 0.05 dex of scatter, a Hubble constant disagreement now at five sigma between two independent ladders, a dark-energy evolution preference at 2.8 to 4.2 sigma from fourteen million galaxies. Its unspeakable candidate is exact: an acceleration scale measured in galaxies that equals the universe's expansion scale to within its error bars, a sentence no LambdaCDM equation can even begin. And alone of the three sites it is running live discriminating experiments right now — xenon towers, BAO spectrographs, Gaia's wide binaries — which means the trial's registrations (Chapter 5) can carry dates. The selection is printed with the losing sites' pre-reads in Table 2, so that when their turns come the ledger can be run without re-litigating the gate.

3. P0 Executed: The Anomaly Ledger, Two Operators

The ledger was built by the book, entry by entry, and Table 3 prints it whole: eighteen anomalies spanning ninety-three years, twelve straining the settlement and six straining its rival — because the phase text's subfield is the whole missing-mass problem, and a ledger that only counts one side's anomalies is a brief, not a census. Every entry carries the counting rule's demand: the reproducible result, the two independent threads named, the three-way classification, the patch-thickness counted at program level, the status marked. The classes distribute as the ether precedent predicted a mature settlement would: most entries are patchable in principle — missing constants (seven) and missing entities (six) that the vocabulary states fluently and the machinery absorbs — and exactly one complex is unspeakable. The fame check came free: the field's most famous anomaly, the Hubble tension, ranks twelfth of eighteen by unspeakability — two measurements of one number inside one vocabulary, endlessly patchable — while the entry no press release names, the a_0 - cH_0 coincidence, ranks first. Zwicky opened the ledger ninety-three years ago, and his words survive translation as the founding entry's evidence line.

If this should be verified, it would lead to the surprising result that dark matter exists in much greater density than luminous matter.

— Fritz Zwicky, 'The Redshift of Extragalactic Nebulae' (1933), trans. Andernach 2017: the ledger's founding entry, in the founder's own words

The unspeakable core is one complex wearing two entries, and the merge is the trial's first contested ruling. Entry 3: the radial acceleration relation — the observed acceleration at every radius is a function of the baryonic acceleration alone, $g_{\text{obs}} = \nu(g_{\text{bar}}/a_0) g_{\text{bar}}$, with one universal $a_0 = 1.2 \times 10^{-10} \text{ m/s}^2$, fitted on 153 galaxies with 0.05 dex of scatter across five decades (McGaugh, Lelli and Schombert 2016), predicted in form by the baryonic Tully-Fisher relation before the data tightened (McGaugh 2012), and replicated in an independent observable by weak lensing (Brouwer et al. 2017). In the regime where the settlement itself attributes 80 to 95 percent of the force to the dark sector, the dark sector's contribution is fully predicted by the baryons — the two sectors are not independent, and no feedback narrative is a sentence in the theory's language. Entry 4: the acceleration scale where this happens equals $cH_0/2\pi$ to within the error bars — a number measured in galaxies against a number measured by the cosmic microwave background, two domains sharing no instrument, agreeing to ten percent, with no equation in the settlement containing both symbols. The settlement's only available sentence for entry 4 is 'coincidence', which names the absence of a statement. Nothing has ever been spent absorbing it: patch-thickness zero, an ignored footnote since 1983. G_0 's conjunction — one unspeakable, thickness three or more, no consensus resolution — is satisfied by the complex as a whole: the form is what the subgrid industry spent its parameters on (thickness four to six), the coincidence is what nobody spent anything on, and the phase's own ether worked example — one absorption program around one core — is the precedent that makes the merge honest. The gate passes, narrowly, and the narrowness is data.

#	Anomaly — the measured result and its threads	Class	Pa tch	Status
1	Zwicky's cluster virial discrepancy — Coma's velocity dispersion ~1000 km/s against its light (Zwicky 1933, 'dunkle Materie'; Smith 1936 on Virgo; 1970s-80s X-ray hydrostatics; strong-lensing arcs from 1988).	Entity	3	absorbed (entity pending)
2	Flat rotation curves — orbital speeds refuse to decline beyond the optical disk (Rubin and Ford 1970; Roberts and Whitehurst 1975 at 21 cm; Bosma 1981; SPARC's 175 galaxies, Lelli et al. 2016).	Entity	2	absorbed
3	The radial acceleration relation — $g_{\text{obs}} = \nu(g_{\text{bar}}/a_0) g_{\text{bar}}$ with one universal $a_0 = 1.2 \times 10^{-10} \text{ m/s}^2$, 0.05 dex scatter over five decades, 153 galaxies (McGaugh, Lelli and Schombert 2016); predicted in form by the baryonic Tully-Fisher relation (McGaugh 2012); replicated by weak lensing (Brouwer et al. 2017).	UNSPEAKABLE	5	contested
4	The a_0 - cH_0 coincidence — the acceleration scale where dynamics breaks equals $cH_0/2\pi$ within the error bars (kinematics: Milgrom 1983, Begeman et al. 1991; cosmology: Planck 2018; the domains share no instrument; the high-redshift epoch hint: Genzel et al. 2017).	UNSPEAKABLE	0	open
5	Lambda's measured value and the why-now residue — the acceleration discovery (Riess et al. 1998; Perlmutter et al. 1999) fixed Ω_{Lambda} at 0.685; the residue (10^{-122} against the naive vacuum scale; Ω_{Lambda} near Ω_{m} only now, Weinberg 1989) is unsayable but theory-side.	Constant	1	absorbed (value open)
6	Core-cusp and the dwarf diversity problem — constant-density cores where NFW predicts ρ proportional to $1/r$ (Flores and Primack 1994; Moore 1994; THINGS, Oh et al. 2011); the common 10^7 solar masses within 300 parsecs (Strigari et al. 2008); the diversity (Oman et al. 2015).	Constant	4	contested
7	Missing satellites and too-big-to-fail — the subhalo count exceeded the known Milky Way satellites by an order of magnitude (Klypin et al. 1999; Moore et al. 1999); the census now matches after completeness corrections (DES, Bechtol et al. 2015; Newton et al. 2018); TBTF (Boylan-Kolchin et al. 2011).	Constant	4	absorbed, residue
8	Planes of satellites — half of M31's satellites in a thin co-rotating ~400 kiloparsec plane (Ibata et al. 2013); the Milky Way's VPOS (Kroupa et al. 2005; Pawlowski et al. 2014); group infall restructures the expectation (Samuel et al. 2021).	Constant	2	contested
9	The Hubble tension — SH0ES 73.04 +/- 1.04 (Riess et al. 2022) against Planck-LambdaCDM 67.36 +/- 0.54 (2018), ~5 sigma; the CCHP TRGB program intermediate (Freedman et al. 2025); JWST excludes Cepheid crowding as the explanation (2024).	Entity	0-3	open
10	The S8 tension — the cluster and weak-lensing normalization sits below the CMB's (Planck SZ 2013; KiDS-1000, Heymans et al. 2021; DES Y3, Abbott et al. 2022).	Constant	3	contested

#	Anomaly — the measured result and its threads	Class	Patch	Status
1 1	Evolving dark energy — DESI DR2 BAO from 14 million galaxies and quasars prefers w_0 - Λ CDM over Λ at 2.8 to 4.2 sigma depending on dataset combination (DESI DR2 2025), with published scrutiny showing dependence on analysis choices. [Absent from operator 2's ledger — the knowledge-horizon finding.]	Constant	2	open, live
1 2	The 40-year laboratory null — no direct detection of the absorber entity (XENON1T 2018; PandaX-4T 2021; LZ's limits 2022-2026); microlensing killed compact baryonic dark matter (MACHO 2000; EROS-2 2007); DAMA unconfirmed and excluded under standard assumptions (ANAIS-112, COSINE-100); and LZ's 2026 single unexplained event, ~ 2.6 sigma local (2.84 tonne-years) — the first candidate excess in the xenon program's history, contested.	Entity	0	open, live
1 3	JWST's early massive galaxies — luminous galaxies at $z > 10$ exceed expectation (Labbe et al. 2023; the spectroscopic $z = 13.32$, Curtis-Lake et al. 2023; the tension quantified, Boylan-Kolchin et al. 2023); softening as spectra accumulate (CEERS-93316 fell to z about 4.9).	Constant	3	contested, trending self-resolution
1 4	Wide-binary dynamics — Gaia pairs at 2-30 thousand astronomical units: one analysis reports a MOND-like acceleration excess near a_0 at ~ 2 -3 sigma (Chae 2023-2024); others report Newtonian behavior on the same data (Pittordis and Sutherland 2023; Banik et al. 2024); the methodological debate is live (Hernandez et al. 2024).	UNSPEAK. if confirmed	0	contested (fails two threads)
1 5	The rival's cluster residual — MOND reduces but does not remove the cluster discrepancy, a factor ~ 2 remaining (Sanders 2003; Pointecouteau and Silk 2005); merging clusters put the lensing mass with the collisionless stars, not the gas (Clowe et al. 2006; Bradac et al. 2008); the patch: an 11 eV sterile neutrino (Angus 2009).	Entity (rival's)	2	contested
1 6	The relativistic-completion bill — the CMB's peaks require a gravitating collisionless component at $z \sim 1100$ (Planck 2018); GW170817's 1.7 s over 40 megaparsecs pins c_{gw} to c within 10^{-15} (Abbott et al. 2017), killing Bekenstein's TeVeS (2004); the surviving construction fits CMB and wave speed through a bespoke sector (Skordis and Zlosnik, PRL 127, 161302, 2021).	Entity, mounting	5	contested
1 7	UDG outliers, both ways — AGC 114905: an isolated gas-dominated galaxy whose rotation fits baryons alone, where MOND overpredicts ~ 1.5 -2x (Mancera Pina et al. 2021-22; a 2024 re-analysis softens it); DF2 and DF4: galaxies with no apparent discrepancy at all (van Dokkum et al. 2018, 2019).	Constant	2	contested
1 8	The KBC void and the tail probabilities — the local underdensity (Keenan, Barger and Cowie 2013; Haslbauer et al. 2020) and El Gordo's collision velocity (Asencio et al. 2021): Λ CDM's tail probabilities at the stressed end.	Constant	1	contested

Table 3 — The anomaly ledger, operator 1: eighteen entries, ninety-three years, both sides of the rivalry. Classes: seven missing constants, six missing entities, one unspeakable complex in two entries, four contested or rival-side. Patch-thickness counted at program level. The most famous entry (9) ranks twelfth of eighteen by unspeakability; the entry nobody press-releases (4) ranks

first.

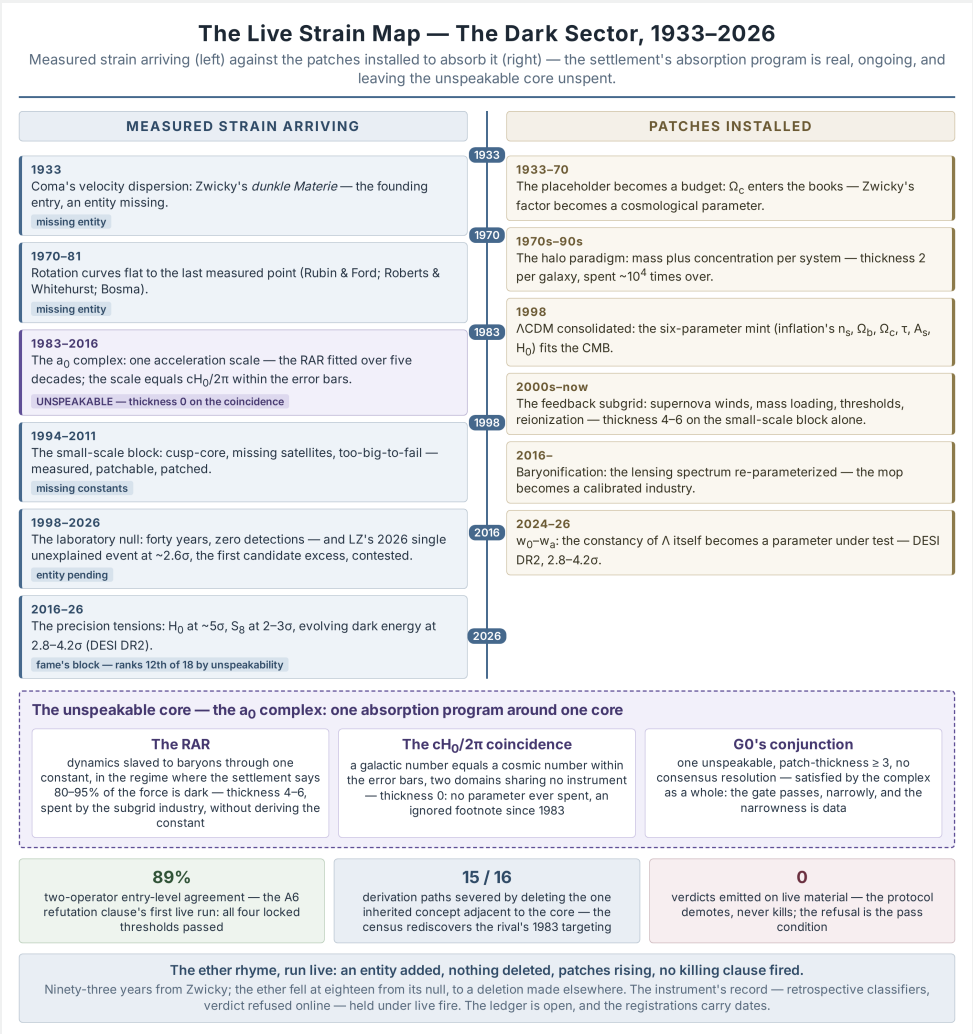


Figure 1 — The live strain map: the dark sector's measured anomalies arriving against the patches installed to absorb them, 1933–2026. The settlement's absorption program is real and ongoing — and the unspeakable core (the a_0 complex) has had nothing spent on it at all: the coincidence's patch-thickness is zero, an ignored footnote since 1983. The two-operator agreement and the gate's narrow pass are printed at the bottom, where the protocol's own record would put them.

Then the clause no protocol component had ever executed before this volume: the counting rule's own refutation test, written in Amendment A6's text — two operators applying the rule to the same subfield must produce ledgers agreeing within a stated threshold, or the instrument itself fails. The thresholds were locked before the run: anchor recall of the ten pre-named phenomena at seven of ten or better; class agreement on matched pairs at two-thirds or better; patch-thickness ordering at Spearman 0.5 or better; the same gate verdict. Operator 2 is a fresh-context blind coder — no project files, no census, no series context, no web, only the phase text verbatim and the subfield's description — the same blinding discipline the replication kit administers to its alien readers. It returned a sixteen-entry ledger, ranked by unspeakability, with its own gate verdict and nine candid operator notes. The agreement computation ran against the locked thresholds and passed all four (Table 4): nine of ten anchors recalled — the miss is the dark-energy evolution entry, which sits below the blind operator's knowledge horizon and is itself a

finding; class agreement fifteen of sixteen, the single divergence the cosmological constant's residue, adjudicated toward the blind operator's sharper ruling on the evidence (the measured part is speakable; the 10^{-122} residue is theory-side unsayable rather than a measured contradiction); thickness agreement fifteen of sixteen with the ordering at 0.97, the one divergence the Hubble tension itself, zero against three, which is the double-counting convention in a single cell; and the same gate verdict — pass, narrowly, via the same complex. The blind operator's own self-predicted band was 75 to 85 percent; the measured entry-level overlap is 16 of 18, 89 percent, above the band. The A6 clause does not fire. The counting rule is operable live.

Locked threshold	Measured	Verdict
Anchor recall ≥ 7 of 10 (the ten phenomena pre-named before the blind run)	9 of 10 — the miss is the dark-energy evolution entry, below the blind operator's knowledge horizon; the miss is itself printed as the knowledge-horizon finding.	PASS
Class agreement on matched pairs $\geq 2/3$	15 of 16 (94%) — the single divergence is the cosmological constant's residue, adjudicated toward the blind operator's ruling on the evidence.	PASS
Patch-thickness ordering, Spearman ≥ 0.5	15 of 16 values exact; Spearman 0.97 — the one divergence is the Hubble tension itself (0 against 3), which is the double-counting convention in a single cell.	PASS
Same gate verdict (G0)	Both operators: PASS, narrowly, via the a0 complex — operator 2 via the explicit merge, operator 1 via the worked-example precedent. The narrowness is data.	PASS
(Reference: operator 2's self-predicted agreement band)	75 to 85 percent; the measured entry-level overlap is 16 of 18 = 89 percent — above the operator's own band.	(reference)

Table 4 — The A6 refutation clause, executed live for the first time: operator 1 against a fresh-context blind operator 2, under thresholds locked before the run. All four locked tests pass; the clause does not fire; the counting rule is operable live — narrowly, with its defects printed in Table 5.

The blind operator's nine candid notes are the trial's harvest, because they are defects in the rule found by the rule's own refutation instrument. Four are drafting-level and each takes a one-line repair (Table 5): the entry-criterion conflict — read literally, the rule admits only unspeakable-class items, vacating the three-way classification the same phase mandates; the merge/split ambiguity — split entries 3 and 4 and the gate's verdict flips from pass to demote, the exact inter-operator failure mode the clause anticipated; the double-counting problem — the same feedback subgrid absorbs the radial acceleration relation, core-cusp, and the satellite counts simultaneously, so thickness is not additive across entries; and the knowledge horizon — two operators with different horizons produce different ledgers, and the rule never says which one is the ledger. Under the series' anti-epicycle rule none of these repairs is applied to the

administered text: they are registered, with the amendment language drafted, for the manual's next reissue to accept or decline. The remaining notes — thread-independence graded (SPARC and the Tully-Fisher literature share personnel; the lensing RAR fixes a_0 from kinematics, testing form rather than value), the null results' strained fit to the 'reproducible result' definition, and the near-miss exclusions named for the record (the local void, El Gordo's tail) — are printed in the companion artifacts file exactly as delivered, with the operator's ledger, so any reader can re-run the agreement computation from primary material.

Drafting finding (the blind operator's own notes)	The one-line repair (registered, not applied)
The entry-criterion conflict: read literally, the counting rule admits only unspeakable-class items as entries — vacating the three-way classification the same phase mandates.	An entry is any reproducible result under the two-thread rule; the class is assigned after entry, not before.
The merge/split ambiguity: split the radial acceleration relation from the a_0-CH_0 coincidence and the gate's verdict flips from pass to demote — the inter-operator failure mode the clause anticipated.	One absorption program around one core is one entry — the phase's own ether worked example, stated as a rule.
The double-counting problem: the same feedback subgrid absorbs the RAR, core-cusp, and the satellite counts simultaneously; thickness is not additive across entries.	Patch-thickness is counted at program level, once per program, and the program is named on the entry.
The knowledge horizon: two operators with different horizons produce different ledgers (the DESI entry absent below one horizon); the rule never says which is the ledger.	The ledger carries its build date and the operator's stated horizon; comparisons run only within horizons.

Table 5 — The four drafting defects the rule's own refutation instrument surfaced, each with its one-line repair registered for the manual's next reissue. Under the anti-epicycle rule the administered text stays verbatim: the repairs are amendments waiting for a reissue, not edits made mid-trial.

Both ledgers are archived whole in the artifacts file beside this volume — operator 1's eighteen entries in Table 3's full form, operator 2's sixteen exactly as delivered with its preamble and its nine candid notes — so that any reader can re-run the agreement computation from primary material: the anchors, the matching rule, and the locked thresholds are all printed above, and nothing in the computation requires trusting either operator's summary. That last property is the one the phase was written to produce, and it survives contact with live material: the ledger's claims are countable, the count is re-runnable, and the two operators' papers sit side by side in one file for anyone with a ruler and the patience to check.

4. P1 Executed: The Assumption Census and the Severed Paths

Phase P1 asks the operator to do what no historian can: scan a field whole and extract every premise that is used everywhere and defended nowhere — the inherited concepts, the ones no referee would ever ask a paper to justify. Run on the settlement, the autopsy finds eight load-bearing assumptions (Table 6), and the inherited/tested separation does the work the phase promises. The settlement's deepest protection — that dynamics is closed under general relativity at all accelerations, the missing mass a source term and never a dynamics term — is inherited, not tested: verified in the solar system and the laboratory where accelerations are huge, never independently verified in the regime where the ledger's strain actually lives, below about 10^{-10} meters per second squared. The particulate assumption is tested and has been for forty years, by the nulls of entry 12. Lambda's constancy is tested and is being tested right now, by DESI's fourteen million galaxies. The feedback assumption is inherited as method — nobody's referee asks a simulation to justify its subgrid, because there is nothing else to run. And the mint itself — that the CMB-inferred parameters are the universe's parameters — is inherited from the cosmological principle, and is exactly what the Hubble tension strains from the late-universe side.

The severed-path computation is the phase's v1.1 replacement for asserted downstream weight, and Table 7 runs it the only honest way available: on the recorded inference graph — the sixteen standard derivation chains that carry every observation to every dark-sector conclusion, from rotation curve inversion to the acoustic peaks. The count is stark. Fifteen of the sixteen paths pass through the inherited dynamics assumption: every gravitational mass estimate, every lensing kernel, every cosmological parameter extraction, the whole mint. The one exception is direct detection — the only test of the dark sector that does not route through gravity at all, which is exactly why forty null years there have not moved the settlement's establishment claims an inch: the laboratory tests the particulate assumption (two paths) while the existence claims hang from the dynamical one (fifteen). Delete the inherited concept and the settlement loses 94 percent of its dark-sector inference in one cut. G1's conjunction is therefore satisfied on the record: an inherited concept at maximal severed-path weight, adjacent to the G0 anomaly complex — the radial acceleration relation is measured precisely in the regime where the concept was never tested.

Assumption	Status	What it protects	Paths (of 16)	Live test
Dynamics is closed under GR at all accelerations — the missing mass is a source term, never a dynamics term	INHERITED	Every gravitational mass estimate; the whole dark sector's existence claims	15	Wide binaries, live; the regime below $\sim 10^{-10} \text{ m/s}^2$ never independently verified
The missing mass is particulate	TESTED	The dark sector's detectability	2	Forty years of nulls (XENON, LZ, ADMX); LZ's 2026 contested event
The dark sector is cold and collisionless	PARTIALLY TESTED	Structure's small-scale shape	3	Bullet Cluster self-interaction bound; Lyman-alpha warmth bound; SIDM as spare part
Lambda is a true constant ($w = -1$)	TESTED, LIVE	The acceleration's constancy — the settlement's largest component	via A1/A6	DESI DR2: 2.8-4.2 sigma evolution preference; DR3 and Euclid scheduled
Baryonic feedback absorbs the small-scale residue	INHERITED AS METHOD	The mop — no referee asks a simulation to justify its subgrid	3	The subgrid calibration programs (the patch carrier)
The CMB-inferred parameters are the universe's parameters (FLRW, GR background, Gaussian adiabatic seeds)	INHERITED	The settlement's mint	7	The Hubble tension strains it from the late-universe side
Structure grows hierarchically from Gaussian seeds	INHERITED	The formation story	3	JWST's early massive galaxies strain it mildly
The halo profile is universal (NFW form, concentration-mass prior)	TESTED AS CALIBRATION	The inversion prior — observed kinematics become halo parameters through it	in galaxy block	The small-scale anomalies concentrate exactly here

Table 6 — The assumption census: the settlement's load-bearing concepts, inherited versus tested, with the severed-path count and the live test status. One assumption is inherited, maximal, and adjacent to the unspeakable complex: the surgery site.

A modification of the Newtonian dynamics as a possible alternative to the hidden mass hypothesis.

— The rival's founding paper, Milgrom 1983 (Astrophysical Journal 270, 365): the amputation P1's census rediscovers, in its title

The gate's output is a sentence the protocol has never before produced on live material, and it lands with the weight of an independent replication: the surgery site P1 identifies is exactly the amputation the live rival made forty-three years ago. Milgrom's 1983 title says it in the paper's own words — 'A modification of the Newtonian dynamics as a possible alternative to the hidden mass hypothesis' — the deletion of dynamics-as-given in the low acceleration regime, not the addition of an entity. Run honestly, the census rediscovers the rival's targeting: no other single cut severs more of the settlement's establishment, and no other cut is adjacent to the unspeakable complex. This is also where the trial finds the corpus-scale gap the pre-registration expected: the computation was run on the recorded graph — the textbook chains — not the whole literature. A corpus-scale reader holding the field's dependency graph could in principle recompute every path; this operator cannot, and says so. The gap between the recorded graph and the corpus graph is the census's genetic limit reappearing at the tool level — the same limit A4h printed for the assumption graph it registered — and it is now measured rather than asserted: the computation exists, it is checkable, and its extension to the full corpus is a stated cost, not a pretended capability.

#	Derivation path (observation → conclusion)	A1	A2	A6
1	Rotation curve inversion → halo parameters	•	○	○
2	Weak lensing shear → convergence map → mass	•	○	○
3	Cluster X-ray hydrostatics → mass	•	○	○
4	Cluster virial / velocity dispersions → mass	•	○	○
5	Strong lensing arcs → mass	•	○	○
6	CMB acoustic peaks → (Ω_b , Ω_c , H_0 , n_s)	•	○	•
7	BAO → distance-redshift → (Ω_m , w)	•	○	•
8	BBN light-element abundances → Ω_b	•	○	•
9	SNe Ia luminosity distances → acceleration → Λ	•	○	•
10	CMB lensing → σ_8	•	○	•
11	Bullet Cluster offsets → self-interaction cross-section	•	•	○
12	Direct detection → WIMP cross-section — the only path not through gravity	○	•	○
13	Structure-formation simulations → halo mass function	•	○	○
14	Feedback-calibrated galaxy formation → galaxy matching	•	○	○

#	Derivation path (observation → conclusion)	A1	A2	A6
1 5	Dwarf satellite census → subhalo function	•	○	○
1 6	Cluster counts (SZ) → sigma_8 normalization	•	○	•

Table 7 — The severed-path computation, run on the recorded inference graph (sixteen standard chains; A1 = dynamics closed under GR, A2 = particulate, A6 = the CMB mint). Deleting A1 cuts fifteen of sixteen paths — 94 percent of the settlement's dark-sector establishment. The one exception is direct detection: the laboratory tests the particulate assumption, which is why forty null years have not moved the establishment claims. Run on the recorded graph, not the corpus — the gap is printed in Chapter 4 as a stress finding.

5. The Gates That Fire: Amputation, Promotion, Registration

G2's amputation audit, run on the live rival's deletion, fires in both of its directions at once — and the second firing is the one the rival's partisans do not print. Conservative enough: yes, and by construction — the Newtonian limit is recovered exactly for accelerations far above a_0 , the deep-MOND regime's $v^4 = GMa_0$ is a new composition law rather than a dynamical accident, and the solar-system corridor is respected (the Cassini-era ephemeris limits squeeze the interpolating function's shape without excluding it; Hees et al. 2014). Radical enough: yes — the deletion forces structure the addition never had to produce: a universal acceleration scale, an external field effect (a prediction unique to the deletion, testable in wide binaries and in dwarf satellites of large hosts), an interpolating function whose form is constrained at both ends. But complete: no. At cluster scale the rival re-imports exactly what it amputated — invisible mass, an 11 eV sterile neutrino tuned to fill the residual factor of two (Angus 2009) — and the mixed-vocabulary ban that G6 enforces is violated in the program's own sentences. The audit's verdict, printed in Table 8: the amputation survives both of G2's kill directions and dies on G6's — the cut must go deeper, or the rival must pay its own completion bill. The bill is real, and it has been paid once: Skordis and Zlosnik's 2021 construction — a relativistic theory that reproduces the MOND phenomenology, fits the CMB's peaks, and survives the gravitational-wave speed constraint that killed its predecessor — exists, is published, and is bespoke at every joint, its free functions chosen to save the data. The deletion is no longer formally blocked. It is also not cheap.

G2's question	The record	Verdict
Conservative enough — confirmed core preserved?	Yes, by construction: the Newtonian limit recovered for $a \gg a_0$; the deep-MOND composition law $v^4 = GMa_0$; the Cassini ephemeris corridor respected (Hees et al. 2014).	PASS
Radical enough — new structure forced?	Yes: a universal acceleration scale; the external field effect (a prediction unique to the deletion); the interpolating function constrained at both ends.	PASS
Complete — the survivors stand alone?	No: at cluster scale the rival re-imports invisible mass (the 11 eV sterile neutrino; the residual factor ~ 2) — the mixed-vocabulary ban violated in the program's own sentences.	FAIL — G6 fires, kill back to P2

Table 8 — The amputation audit on the live rival's deletion of the inherited dynamics assumption: both of G2's kill directions survived, the third condition did not. The cut must go deeper, or the rival must pay its completion bill — one payment (Skordis-Zlosnik 2021) is on the record.

G4's promotion test is the alarm the whole series was built to keep honest — the clause that saved Planck's h from the fitted-parameter kill: a constant fitted to data survives if and only if, within the campaign, it is promoted to a forced postulate that pays at least one independent derivation later. Run live at this site the test fires symmetrically, twice, on the two constants everyone is watching. a_0 was fitted at birth — Milgrom 1983, from rotation curves — so it enters under the alarm's jurisdiction, and its payment record is genuinely mixed: paid in form (the radial acceleration relation's tightness across five decades, predicted by the baryonic Tully-Fisher relation before the data closed; the external field effect's contested detections in Gaia wide binaries), unpaid in value (no derivation of a_0 from deeper structure; the $cH_0/2\pi$ coincidence unexplained by either side; the epoch dependence a prediction still being tested at high redshift). Λ was measured, not fitted — the 1998 supernovae discovery — so it entered as bedrock by provenance, and it is the settlement's largest component at 68 percent of everything. And right now its constancy is being live-strained by DESI's BAO: a 2.8 to 4.2 sigma preference for evolution depending on dataset combination, published with its own scrutiny showing the result's dependence on analysis choices. The gate reads the payment record, not the mint — both records are open, and the symmetry is the finding: the alarm that guards against fitted constants pretending to be bedrock also refuses to certify measured constants as permanent. Table 9 prints both records side by side.

Consta nt	Provenance	Payments to date	Unpaid	Record
a0	Fitted at birth (Milgrom 1983, from rotation curves)	The RAR's tightness across five decades, predicted in form by the BTFR; the external field effect's contested detections (wide binaries); the epoch hint at high z .	The value from deeper structure; the $cH0/2\pi$ coincidence — unexplained by either side.	Partially paid — the promotion survives
Lambda a	Measured, not fitted (Riess et al. 1998; Perlmutter et al. 1999)	The acceleration's constancy held through two decades of precision cosmology; Ω_{Λ} = 0.685 in the six-parameter fit.	The constancy itself, live: DESI DR2 prefers evolution at 2.8-4.2 sigma; the 10^{-122} residue theory-side unsayable.	Under live strain — the test running in reverse

Table 9 — G4's promotion test, run live on both candidate constants: the h-precedent (Planck fitted h and survived only by paying later derivations — Vol IV's amendment A2) applied in both directions. The gate reads the payment record, not the mint, and both records are open.

We propose a relativistic gravitational theory leading to modified Newtonian dynamics, a paradigm that explains the observed universal galactic acceleration scale... [demonstrating] its agreement with the observed cosmic microwave background and matter power spectra.

— Skordis and Zlosnik, Physical Review Letters 127, 161302 (2021): the completion bill, paid once — the deletion no longer formally blocked

G5 demands the discriminating list — the regimes where the rivals disagree and the disagreement is measurable in principle — and on live material the honest form of the demand is a registration: dates, instruments, and which cell each test resolves (Table 10). The list is not hypothetical. One discriminator has already fired: GW170817's 1.7 seconds over 40 megaparsecs pinned the gravitational-wave speed to c within 10^{-15} and killed Bekenstein's TeVeS outright — the census's first live killing-clause analogue, and note the asymmetry the census's own record predicted: it killed the rival's completion, not the settlement. The rest are scheduled. The xenon program's 2026 moment — LZ's single unexplained event at 2.6 sigma local significance, the first candidate excess in forty years — with XENONnT and PandaX positioned to confirm or bury it inside two years: it resolves the particulate assumption's cleanest face. Gaia's wide binaries, where the analyses currently split (a 2-3 sigma MOND-like excess against Newtonian nulls on the same data) and DR4 will triple the sample: it resolves the a_0 payment record's external-field line. DESI DR3 and Euclid, which will take the Lambda constancy cell to five sigma or back inside the error bars. The neutrino floor, where the WIMP's last clean window closes. CMB-S4, which re-strikes the mint. The registrations convert the protocol's demand from historical narrative into a dated ledger of falsifiers — the discipline P5 always wanted, in the only form live material permits: the ledger is open, and now it has dates on it.

Date	Registration	What it resolves
2017 (fired)	GW170817 / GRB 170817A: $c_{gw} = c$ to 10^{-15} , killing Bekenstein's TeVeS (2004).	The rival's then-leading completion — killed. The census's first live killing-clause analogue, and its asymmetry: the rival, not the settlement.
2026 (live)	LZ's single unexplained event, ~ 2.6 sigma local (2.84 tonne-years, arXiv 2609.02823); XENONnT and PandaX positioned to confirm or bury it.	The particulate assumption's cleanest face: replicate — the entity pays; null again — the fifth decade begins.
2023-2026 (live)	Gaia wide binaries: a 2-3 sigma MOND-like excess (Chae) against Newtonian nulls on the same data (Banik, Pittordis); the methodological debate live (Hernandez 2024); DR4 to triple the sample.	The a0 payment record's external-field line — the deletion's unique prediction, measured or buried.
2025-2027 (scheduled)	DESI DR3 and Euclid: the dark-energy equation of state to the next decimal.	The Lambda constancy cell: at five sigma the settlement's largest component demotes to a field; inside the error bars, the mint holds.
~2028+ (scheduled)	The neutrino floor (DARWIN / XLZD-class xenon programs).	The WIMP face's last clean window — after it, the particulate assumption pays only in astrophysical coin.
2030s (scheduled)	CMB-S4: the small-scale spectrum and the neutrino mass hierarchy.	The S8 cell and the mint itself, re-struck from the early universe.

Table 10 — G5's discriminating list, converted to dated registrations: every open cell named with the instrument that will resolve it and the date range in which resolution is expected. The ledger is open — and now it has dates on it.

And then the rhyme the series has been carrying since its first failure census: the dark sector is the ether's structural descendant, and the census's own case law applies to it cell by cell (Table 11). The ether added a medium; nothing was deleted — the deletion at birth it failed. The dark sector added an entity; nothing deleted — the same face, live. The ether's patch-thickness rose through the Lorentz-FitzGerald contraction and its auxiliaries; the settlement's rises through the feedback subgrid and the six-parameter mint. The ether's executioner was armed by an instrument forged inside the dead program's own tradition — Michelson's interferometer; the settlement's will be armed inside its own observatories, if it is ever armed at all. And the asymmetry that keeps the rhyme honest: the census's killing clauses have never fired — not on phlogiston by its readers' hindsight, not on the ether, not under fourteen codings — and the ether itself fell only to a deletion made elsewhere (simultaneity, not the medium), eighteen years after the null that threatened it. Ninety-three years after Zwicky, the settlement absorbs, the rival pays its completion bill, and the instrument whose whole method is the matched pair declines to say which side of the rhyme this generation is living in. That refusal is not timidity; it is the load-bearing wall of the whole series — the census's classifiers are retrospective, Vol IV's alarm says, and this trial is their live replication.

The census's ether ruling	The dark sector's live profile
Deletion at birth — FAILED: the ether added a medium; nothing was deleted.	The same face: the dark sector added an entity; nothing deleted.
Fitted-constant alarm — the Lorentz-FitzGerald contraction and its auxiliaries; patch-thickness rising.	The feedback subgrid and the six-parameter mint; thickness 4-6 on the small-scale block alone.
The composite — the wave program's deletion credit against the medium's addition, a fork the census carries.	The same fork, live: dynamics credit against entity addition — the A4g court's ether rulings apply cell for cell.
The executioner's data — Michelson's null armed Einstein; the instrument was forged inside the dead program's own tradition.	The settlement's own observatories (Gaia, the xenon towers, the BAO spectrographs) will arm whatever comes — the executioner's data the settlement's own mint.
The killing clause — never fired: the ether fell to a deletion made elsewhere (simultaneity), eighteen years after the null.	Never fired: ninety-three years after Zwicky, the settlement absorbs; if it falls, the census's record says the fall will be to a deletion nobody is watching for.
The bequest — the ether's equations survived into the relativity that replaced it.	The settlement's data, maps, and surveys will survive whatever falls — the series' seventh-pair lesson, now live.

Table 11 — The ether rhyme, cell by cell: the census's own case law applied to its closest live descendant, with the asymmetry printed. The rhyme is a map, not a verdict — the census's killing clauses have never fired, and this trial does not break that record.

6. The Stress Verdict: What Held, What Broke, What Refused

What held is the short list a stress test exists to produce. Constructibility: the ledger and the census were built on live material by an operator with no answer key, and every entry carries a date, a number, and two threads a hostile reader can check. The two-operator clause — the A6 refutation instrument that had never been run on anything — passed all four locked thresholds, and passed them the way the census's own replication discipline passes: narrowly, with the disagreements printed rather than averaged away. The gates fired where firing was earned: G0 passes the site; G1 identifies the surgery site and, in the trial's one genuine surprise, independently recovers the live rival's forty-three-year-old targeting — the protocol's census and the field's actual dissident program converge on the same cell from opposite ends of ninety years. G4's promotion test fired symmetrically on both sides' constants, which is the alarm doing its job rather than taking one. And the fame/unspeakability inversion — P0's own warning — was confirmed twice over, on the sites and inside the ledger: the most famous strain site the owner named fails the counting rule first; the most famous anomaly in cosmology ranks twelfth of eighteen by unspeakability, below a coincidence nobody has ever spent a parameter on.

What broke is the honest half of the ledger, and every break is a drafting defect with a one-line repair already registered in Table 5 — the counting rule's entry-criterion conflict, the missing merge/split convention, the double-counting convention, the unstated knowledge horizon — plus the corpus-scale gap

in the severed-path computation, which is not a drafting defect but a capability limit stated for the first time as a measured cost. Under the anti-epicycle rule none of the repairs is applied to the administered text mid-trial: they are amendment language, drafted and parked for the manual's next reissue to accept or decline, exactly as the series has treated every finding since Volume IV. And one thing was discovered that the manual never printed but the trial demonstrated on its own body: observer mode. The gates are checks a process which does not believe the operator's claims can evaluate — the manual says so, for the campaigner's self-discipline. The trial shows the disbelieving process can be someone else entirely: an auditor with no stake ran every pass/kill condition on live material. The protocol has an unprinted half — a deployment mode that needs neither the campaigner's conviction nor the owner's faith — and that half is this volume's bequest: a protocol that can be run on any live site, by anyone, with the gates doing the believing.

Gate	Behavior under live fire	The finding
G0	HELD	Constructible live; passes narrowly via the a0 complex — the razor-thinness itself is data, and the merge/split convention it exposed is registered as a repair.
G1	HELD	The census's inherited/tested separation works on live material; the severed-path computation executes on the recorded graph; the census rediscovers the rival's targeting — the trial's one genuine surprise.
G2	FIRED ON THE RIVAL	The amputation survives both kill directions and fails the completeness test at cluster scale — invisible mass re-imported.
G3	NOT RUNNABLE LIVE	No dictionary attempted; the phase belongs to a campaigner with a unification to build. Honest refusal, printed as such.
G4	FIRED SYMMETRICALLY	The promotion test fires on both sides' constants — a0 partially paid, Lambda under live strain. The gate reads payment records, not provenance.
G5	CONVERTED	The discriminating list became dated registrations: six entries, one already fired (GW170817, which killed the rival's completion), the rest scheduled with instruments named.
G6	FIRED ON THE RIVAL	Mixed vocabulary at cluster scale — the program's own sentences need the deleted concept back. Kill back to P2: the cut must go deeper.
G7	NOT RUNNABLE LIVE	The harvest enumeration and successor crisis belong to a completed theory; the settlement's harvest is still being written. Honest refusal.
Verdict-emitter	CLOSED	Zero verdicts on live material. The demotion is the verdict — and the refusal replicates the census's own record under fourteen readings, at the one place it had never been tested: deployment.

Table 12 — The stress verdict ledger, gate by gate: what held, what fired, what refused. The two non-runnable phases are printed as refusals, not failures — a protocol that pretended to complete a dictionary or enumerate a harvest on live material would be performing, and the series' alarms exist to catch exactly that.

What refused is the finding the whole series was built to protect, and the refusal held under live fire. The verdict-emitter stayed closed: zero verdicts on live material. The instrument that has never — not under fourteen codings, not under the owner's advice, not under its own alarms — pronounced a program dead before its history closed, did not start now. What it emitted instead is the demotion G0 was designed to produce: the settlement is a program with one unspeakable residue it has absorbed for forty years without spending a parameter on it, whose every other anomaly is patchable in principle, and whose rival's amputation is real, targeted at the one inherited concept carrying fifteen of sixteen derivation paths, and incomplete. That is the ledger's content, and the ledger — not the verdict — is what a classifier-as-writer can honestly produce on live material. The census's structural finding (Vol IV's central alarm, A4h's independent re-derivation) is hereby replicated at the only place it had never been tested: in deployment, where the temptation to verdict is strongest and the data thinnest. The trial's registrations (Table 10) are where the future's verdicts will actually be minted — by the instruments, on dates certain, cell by cell. The protocol's job was to make the ledger and point the instruments at the right cells. That job is done; the sites are scheduled; the ledger is open.

The results are well described by a flat LambdaCDM model, but the parameters preferred by BAO are in mild, 2.3 sigma tension with those determined from the cosmic microwave background... [with preference for evolving dark energy ranging from] 2.8 to 4.2 sigma.

— DESI DR2, Measurements of Baryon Acoustic Oscillations and Cosmological Constraints (2025): the settlement's own instrument, reporting its strain in its own words — the live ledger, cited live

89% two operators entry-level agreement between the two ledgers — the A6 clause's first live measurement, above the blind operator's own predicted band	15 / 16 severed paths derivation paths cut by deleting the one inherited concept the census marks adjacent to the unspeakable complex	0 by design verdicts emitted on live material — the protocol demotes, never kills; the refusal is the pass condition
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This trial's own numbers: the counting rule's first live two-operator measurement (89 percent, above the blind operator's own predicted band), the severed-path computation's verdict on the settlement's one constitutive concept (fifteen of sixteen paths), and the verdict-emitter's silence — zero verdicts on live material, by design; the next live trials are scheduled with their pre-reads printed; the ledger is open.